

23CS502	SOFTWARE TESTING	L	T	P	C
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COURSE OBJECTIVES:

The objective of this course is to:

- 1 To provide students with an understanding of Core Testing concept.
- 2 To learn the functional and non-functional testing.
- 3 To understand the different types of User Acceptance testing and end-to-end testing.
- 4 To get familiarize with the best practices of Testing.

INTRODUCTION TO AUTOMATION TESTING WITH SELENIUM

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What is Software Testing, Why Software Testing, Benefits of Software Testing, Software Test Levels, Unit Testing, Integration Testing, System Testing, Acceptance Testing, Software Test Types, Functional testing, Non-functional testing, Change Related Testing. Test Scenario Design - Functional and non-functional test scenarios, identify and write business critical scenarios. Test Case Design - pre-requisites, test steps and expected results for test cases, Positive and negative test cases for each scenario, Test case prioritization, Test case optimization technique. RTM, DSR. Classes and Objects, Inheritance, and Polymorphism, Exception Handling, Collections, and, Collections (List), JDBC Connectivity, Creating CRUD OPERATION JDBC Connectivity

WORKING WITH SELENIUM

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Selenium webdriver - Maven Configuration, WebDriver Commands, Navigation Command, Selenium locators - Selenium Locators Basics (id, name),Xpath and css locators, Selenium WebElement - Handling of Form Elements, Synchronization Methods, Selenium Exceptions, Keyboard and mouse handling, Alert handling, Iframe Handling in Selenium, Java Script Execution, Handling WebTable and calendar. Introduction to TestNG - TestNg Introduction, Advantage of testNg , testNG Annotations, Test data preparation, Generation of TestNG Reports, Implicit wait and Explicit wait.

TESTING FRAMEWORK

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Testing Frameworks - Data driven testing using Apache POI, POM. Extent Reports - HTML Report Generation using Extent Reports, Attaching Screenshot in HTML Report. Log4j - configuring log4j Property files, Log4j - parameters for Properties file, Log levels and logging using log4j, Hybrid framework implementation., Creating the POM with a Hybrid framework folder structure, Implementing the Hybrid framework in POM.

TOTAL (Theory): 15 PERIODS

Lab Component:

- 1 Develop a program to automate the login process for a specified webpage using Selenium.
- 2 Write an automation script using Selenium to handle form elements on a given website.
- 3 Create an automation script with Selenium to interact with specific web elements on a designated webpage.
- 4 Implement automation scripts using TestNG prioritizing different test cases for efficient

- 5 Develop automation scripts with TestNG, incorporating seven levels of logging for detailed analysis while testing a specified website.
- 6 Execute application tests using designed test cases and generate an HTML report for a comprehensive overview.
- 7 Design and implement a hybrid framework for a ticket booking system, along with associated test cases.
- 8 Develop a hybrid framework and associated test cases for a hotel room booking system.
- 9 Design and develop a hybrid framework and relevant test cases for a hospital appointment application.
- 10 Implement a hybrid framework and associated test cases for an e-commerce application.
- 11 Develop a hybrid framework and design test cases for comprehensive testing of an insurance website.

TOTAL (Lab): 60 PERIODS
TOTAL: 75 PERIODS

COURSE OUTCOMES (COs):

Upon successful completion of this course, students will be able to:

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| CO1 | Plan and apply the appropriate level of testing within the context of a software development application to the satisfaction of its beneficiaries. | [AP] |
| CO2 | Analyze specific and measurable test cases to ensure coverage and traceability to requirements | [A] |
| CO3 | Understand the problem of reporting techniques, metrics, and testing status reports and communicate testing results to colleagues, managers, and end users | [U] |
| CO4 | Apply testing models, processes and practices appropriate for the software development lifecycle model of a project | [AP] |
| CO5 | Apply principles and practices of test-driven development to improve testing quality and reduce delivery times | [AP] |
| CO6 | Inspect the various testing processes towards the continuous delivery of a software product | [A] |

TEXTBOOKS:

1. Rex Allen Jones II, "Absolute Beginner, Part 1 Selenium Webdriver for Functional Automation Testing", 1st Edition, Createspace Independent Pub, 2016
2. S Basu, "Selenium with Python Simplified for Beginners", 1st Edition, 2020
3. Paul Watson, "Selenium webdriver with Node.js: Beginner's Guide", 1st Edition, CreateSpace Independent Publishing Platform, 2016

REFERENCE BOOKS:

1. Satya Avasara, "Selenium Web Driver Practical Guide", 1st Edition, Packt Publishing

Techniques Using the Selenium WebDriver", 1stEdition, Apress, 2020

3. Pinakin Ashok Chaubal, "Selenium Framework Design in Keyword-Driven Testing: Automate Your Test Using Selenium", 1stEdition, BPB Publications, 2020

WEB REFERENCES:

1. <https://www.coursera.org/projects/building-test-automation-framework-using-selenium-and-testng>
2. <https://www.edx.org/professional-certificate/delftx-automated-software-testing>
3. https://onlinecourses.nptel.ac.in/noc22_cs12/preview
4. <https://www.nextgenerationautomation.com/post/selenium-coding-exercises>
5. <https://www.studytonight.com/maven/build-and-test-maven-project>
6. <https://www.tutorialspoint.com/selenium-for-software-testing-getting-started/index.asp>